Dynamic Macroeconomics II

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Fall 2019

A Monetary Model
Second Part

Version: October 21, 2019

Monetary ideas

- We use the model to explain some important ideas about money and public finance:
  1. Quantitative theory of money.
  2. Deficits cause inflation.
  3. Fiscal prerequisites of zero inflation.
  4. Sargent and Wallace's (1981) "unpleasant monetarist arithmetic".
  5. The optimal quantity of money.

Quantitative Theory of Money

What you learned in Monetary Theory and Policy:

- Consider

$$P_t = \frac{v_t M_t}{Y_t}$$

  where P_t is price level, M_t is money supply, v_t is money's velocity of circulation and Y_t is household's real expenditures.

- If velocity of circulation of money and real expenditures are constant and do not depend on the quantity of money, a change in money supply leads to a percent change in the price level of the same magnitude: $\Delta\% M \implies$ same $\Delta\%$ in P.

Quantitative Theory of Money in the shopping model

We use the graphs for long-run and short-run equilibrium.

- Consider two countries with identical parameters $(\tau_0, \tau, g, B, B_0)$ but a different M_0. (Or two stationary eq. that differ only in initial money supply.)
- Country 1's initial monetary base is M_0, while Country 2 has $M_0' = \lambda M_0$, $\lambda > 0$.
- **Important**: note that R_m is equal across countries. R_m does not depend on M_0. Long-run graph is unchanged.
- Now let's look at short-run graph . Note that M_0/p_0 is unchanged!
- The effect is to multiply Country 2's equilibrium price level by λ, leaving all other equilibrium values unaltered.
- $M_0/p_0 = \lambda M_0/p_0' \implies P_0' = \lambda P_0$

Deficits cause inflation

We saw that when we chose to work with 1 of the 2 eq. of the model. The dynamics were:

- Higher deficit needs to be financed with higher seigniorage.
- Higher seigniorage requires more inflation.
- Higher inflation tax more than compensates reduction in demand for real balances.

Fiscal prerequisites of zero inflation

For now we do not ask ourselves which inflation is optimal. We simply ask what fiscal policy is necessary to have zero inflation.

- From government's budget constraint for $t \geq 1$,

$$g - \tau + \frac{B(R-1)}{R} = f(R_m)(1 - R_m)$$

to obtain $\pi = 0$ (inflation *rate* at 0), we need to impose $R_m = 1$:

$$g - \tau + \frac{B(R-1)}{R} = 0$$

or (**check**)

$$B = \frac{R}{R-1}(\tau - g) = \sum_{t=0}^{\infty} R^{-t}(\tau - g)$$

- Assume $B > 0$. Real value of public debt must be equal to present value of net-of-interest government surplus.
- Government renounces to seignorage and has to have a primary surplus.

Unpleasant Monetarist Arithmetic

That is the title of Sargent and Wallace's (1981) classic paper.

- We describe the paradoxical consequence of an open-market operation that reduces money supply.

What we learned in Monetary Theory and Policy:

- Open-market **purchase**: a central bank buys public debt from private banks.
- In exchange, it makes a deposit in the private banks' accounts on the liabilities side of the balance of the central bank.
- The final consequence is an **increase** in money supply.
- Open-market **sale**: a central bank sells public debt to private banks.
- To complete the transaction, central bank makes a withdrawal from the private banks' accounts.
- Final consequence: a **decrease** in money supply.

Open-market operation in the shopping-time model

- Suppose an open-market **sale** at time $t = 0$, defined as a decrease in M_1 along with an increase in B, while fiscal policy variables (τ_0, τ) remain constant.
- We look at its consequences first looking at the long-run graph, then at the short-run graph.
- Long-run graph: effect of this policy can be analyzed by increasing B, which shifts the permanent gross-of-interest deficit upward by $R/(R-1)$ times the increase in B.
- Unambiguously, the stationary inflation rate increases.

- Let's look at short-run graph: there are two effects.
- First, the lower R_m is a downward movement along the curve.
- Second, the higher B shifts the curve upwards.
- The new value for M_0/p_0 could go up or down.
- For class: draw case in which it goes up, which means that p_0 falls.
- Paradox: a lower price level today can be accompanied by permanent higher inflation.
- Sargent & Wallace (1981) called this "unpleasant monetarist arithmetic".
- Intuition: decrease in M_1 should lower inflation. But asset sale implies more resources necessary to pay debt. Therefore inflation goes up.

!

Optimum quantity of money

- Friedman (1969) argued that the marginal cost of printing money is near zero.
- On the other hand, household's marginal cost of holding one more unit of money is nominal interest rate, usually > 0.
- Go over **graphs** drawn in class.
- Equating private and social marginal costs implies making the nominal interest rate equal to zero: Friedman rule.
- This in turn implies negative inflation: deflation.
- On this last point, remember Fisher's equation: $1 + i = (1 + r)(1 + \pi)$.
- Taking logs, then imposing Friedman's we get $\pi = -r < 0$.
- Inflation is equal to the negative of the real interest rate.

Two questions:

1. How to impose Friedman's in shopping-time model?
2. More important: is it optimal? Why? Answer is **yes**

Implementation:

- In terms of the model:

$$\frac{R_m}{R} = \frac{1}{1+i} = 1$$

$$R_m = \frac{1}{\beta} > 1$$

- How to achieve that value?
- By running a sufficiently large gross-of-interest **surplus**

$$g - \tau + \frac{B(R-1)}{R} < 0$$

  the government can attain any value of $R_m \in (1, 1/\beta)$.
- Needs a combination of low spending, low debt, high taxes.
- See graph.

Is this optimal? Intuition:

- Note that consumer prefers eq. with high R_m, as it increases money demand, reduces time spent shopping, increases leisure.
- There is no trade-off with consumption, which is given by $y = c + g$, both y, g are exogenous.
- Therefore it is optimal to supply a "large amount" of **real** money balances, and minimize time spent shopping.

Specifics:

- Recall the equation

$$\frac{R_t - R_{mt}}{R_t}\lambda_t = -\mu_t H_{m/pt}$$

- Previously we have used certain function(s) H.
- On the right-hand side need an extra assumption, otherwise solution is to inject an arbitrary, undetermined, amount of real balances to reduce shopping time H.
- Extra assumption: $\forall c \;\; \exists \; m_{t+1}/p_t = \psi(c)$ such that $H_{m_{t+1}/p_t} = 0$ for $m_{t+1}/p_t \geq \psi(c)$.
- See **graph** representing this assumption.
- To achieve allocation government needs to impose, on left-hand side, $R = 1/\beta = R_m$, which is equivalent to Friedman's.

?

- To put it differently:
  had we set up Ramsey Problem, the solution woud be Friedman's!
- Finally, note that in this solution $p_{t+1}/p_t = \beta < 1$, i.e. there is deflation.

Phelps' critique to Friedman

- Friedman rule's optimality arises when the government uses only lump-sum taxes.
- Need to pay g: Increasing taxes does not distort the economy.
- Is Friedman rule optimal when government raises revenue by means of distorting taxes, as in real life?
- Optimal inflation rate could be strictly positive.
- See Mulligan & Sala-i-Martin (*JMCB*, 1997) for a discussion.
- Optimal inflation rate is small but positive, around 1%.
- What happens with optimal inflation if we modify shopping-time model to include distortionary taxation?

Model

- Includes labor supply n_t. Utility depends on consumption and leisure l_t.
- Technology: $y_t = n_t$
- Feasibility condition: $y_t = c_t + g_t$.
- Household's time constraint: $1 = l_t + s_t + n_t$.
- Shopping technology H is homogenous of degree $\nu \geq 0$ in consumption c_t and real money balances $\widehat{m}_{t+1} = m_{t+1}/p_t$:

$$s_t = H(c_t, \widehat{m}_{t+1}) = c_t^{\nu} H(1, \frac{\widehat{m}_{t+1}}{c_t}) \;\; \forall c_t > 0$$

- By Euler's theorem:

$$\nu H(c, \widehat{m}) = H_c c + H_{\widehat{m}} \widehat{m}$$

- For any consumption level c, there exists a point of satiation in real money balances ψc such that:

$$H_{\widehat{m}}(c, \widehat{m}) = H(c, \widehat{m}) = 0 \;\; \forall \widehat{m} \geq \psi c$$

- Let's make a graph of this function.

- Since real wage is 1, household's income is given by $(1-\tau_t)(1-l_t-s_t)$.
- Sequential budget constraint:

$$c_t + \frac{b_{t+1}}{R_t} + \frac{m_{t+1}}{p_t} = (1-\tau_t)(1-l_t-s_t) + b_t + \frac{m_t}{p_t}$$

- Write budget constraint for time $t+1$ and replace b_{t+1} to get:

$$c_t + \frac{c_{t+1}}{R_t} + \frac{b_{t+2}}{R_t R_{t+1}} + \left(1 - \frac{p_t}{p_{t+1}}\frac{1}{R_t}\right)\frac{m_{t+1}}{p_t} + \frac{m_{t+2}/p_{t+1}}{R_t}$$
$$= (1-\tau_t)(1-l_t-s_t) + \frac{(1-\tau_{t+1})(1-l_{t+1}-s_{t+1})}{R_t} + b_t + \frac{m_t}{p_t}$$

- Recall that

$$1+i_t = \frac{R_t}{R_{mt}} \implies 1 - \frac{p_t}{p_{t+1}}\frac{1}{R_t} = \frac{i_t}{1+i_t}$$

- Arrow-Debreu price: $q_t = \prod_{i=0}^{t-1} R_i^{-1}$, with $q_0 = 1$.
- Transversality conditions:

$$\lim_{T\to\infty} q_T \frac{b_{T+1}}{R_t} = 0$$
$$\lim_{T\to\infty} q_T \widehat{m}_{T+1} = 0$$

- Household's present-value budget constraint (**check**):

$$\sum_{t=0}^{\infty} q_t \left(c_t + \frac{i_t}{1+i_t} \widehat{m}_{t+1} \right) = \sum_{t=0}^{\infty} q_t (1-\tau_t)(1-l_t-s_t) + b_0 + \widehat{m}_0 \tag{1}$$

- Substitute $s_t = H(c_t, \widehat{m}_{t+1})$ into equation (1)

Now we start setting up the Ramsey Problem.

- We want to find out what is the optimal amount of money.
- We will see if solution corresponds to Friedman's, or not.
- Next step for Ramsey Problem: Eliminate prices and taxes from intertemporal constraint.
- To do that we use FOCs of consumer's problem.
- Let λ be Lagrange multiplier on intertemporal budget.

- FOCs:

$$\begin{aligned} c_t &: \beta^t u_{ct} - \lambda q_t[(1-\tau_t)H_{ct}+1] = 0 && (2) \\ l_t &: \beta^t u_{lt} - \lambda q_t(1-\tau_t) = 0 && (3) \\ \widehat{m}_{t+1} &: -\lambda q_t\left[(1-\tau_t)H_{\widehat{m}t+1} + \frac{i_t}{1+i_t}\right] = 0. && (4) \end{aligned}$$

- Combine (2) and (3) to get consumption-leisure equation:

$$\frac{u_{lt}}{1-\tau_t} = u_{ct} - u_{lt}H_{ct}$$

- From equation (3) and $q_0 = 1$, get the Arrow-Debreu price:

$$q_t = \beta^t \frac{u_{lt}}{u_{l0}} \frac{1-\tau_0}{1-\tau_t}$$

- Using equation (4) obtain:

$$\frac{i_t}{1+i_t} = -(1-\tau_t)H_{\widehat{m}t+1}$$

- Suppose $b_0 = m_0 = 0$.
- Eliminate prices and taxes from household's present-value budget constraint by substituting:

$$q_t = \beta^t \frac{u_{lt}}{u_{l0}} \frac{1-\tau_0}{1-\tau_t}$$
$$\frac{i_t}{1+i_t} = -(1-\tau_t) H_{\widehat{m}t+1}.$$

and multiply by

$$\frac{u_{l0}}{1-\tau_0}$$

- Obtain Implementability Condition (**check**):

$$\sum_{t=0}^{\infty} \beta^t \left[\left[u_{ct} - u_{lt} H_{ct} \right] c_t - u_{lt} H_{\widehat{m}t+1} \widehat{m}_{t+1} - u_{lt} (1 - l_t - H(c_t, \widehat{m}_{t+1})) \right] = 0$$

- Use Euler's theorem
  $\nu H(c,\widehat{m}) = H_c c + H_{\widehat{m}}\widehat{m} \implies H_{\widehat{m}}\widehat{m} = \nu H(c,\widehat{m}) - H_c c$
  to get (**do the algebra, check**):

$$\sum_{t=0}^{\infty} \beta^t \left[u_{ct} c_t - u_{lt} \left[1 - l_t - (1-\nu) H(c_t, \widehat{m}_{t+1}) \right] \right] = 0$$

- Ramsey problem: maximize

$$\sum_{t=0}^{\infty} \beta^t u(c_t, l_t)$$

  subject to Implementability Condition (with Lagrange multiplier ϕ) and feasibility $1 - l_t - H(c_t, \widehat{m}_{t+1}) = c_t + g_t$ (with Lagrange multiplier θ_t).

- Set up the Lagrangian and **check** that we get the following first-order conditions. Actually, for main result we only need derivatives with respect to leisure and real money balances

$$
\begin{aligned}
c_t: \quad & u_{ct} + \phi\{u_{cct} c_t + u_{ct} - u_{lct}[1 - l_t - (1-\nu)H(c_t, \widehat{m}_{t+1})] \\
& + (1-\nu)u_{lt}H_{ct}\} - \theta_t[H_{ct} + 1] = 0 \\
l_t: \quad & u_{lt} + \phi\left[u_{clt} c_t + u_{lt} - u_{llt}\left[1 - l_t - (1-\nu)H(c_t, \widehat{m}_{t+1})\right]\right] = \theta_t \\
\widehat{m}_{t+1}: \quad & H_{\widehat{m}t+1}\left[\phi(1-\nu)u_{lt} - \theta_t\right] = 0
\end{aligned}
$$

- Look at the FOC for real money balances.
- It is satisfied if either $H_{\widehat{m}t+1} = 0$ or $\phi(1-\nu)u_{lt} - \theta_t = 0$.
- The second possibility cannot be a solution of the problem.

- Let's prove it.
- By assumption, $\nu \geq 0$.
- **Case** $\nu > 1$**.** Multipliers ϕ and θ_t will either be zero or have opposite signs.
- However, ϕ is always positive and by insatiability of preferences θ_t is strictly positive.
- **Case** $\nu = 1$**.** Implies $\theta_t = 0$, but we just argued that it is strictly positive
- **Case** $\nu \in [0, 1)$**.** Solve for u_{lt} from FOC for real money balances, subsitute into FOC for leisure to get (**check**):

$$u_{lt} + \phi \left[u_{clt} c_t + \nu u_{lt} - u_{llt} \left[1 - l_t - (1 - \nu) H(c_t, \widehat{m}_{t+1}) \right] \right] = 0$$

  which contradicts assumptions $u_c, u_l > 0$, $u_{cc}, u_{ll} < 0$ and $u_{cl} \geq 0$.
- **Result:** When the transaction technology is homogeneous of degree $\nu \geq 0$, the only solution is $H_{\widehat{m}t+1} = 0$.

- This can be accomplished by satiating the economy with real balances.
- The associated monetary policy sustains a zero nominal interest rate.
- I.e. Friedman's Rule is Ramsey-optimal.
- Given the assumption that for every consumption level c there exists a satiation point from real balances $H_{\widehat{m}} = H = 0$, in Ramsey's solution shopping time is 0!
- Note that Friedman's being optimal is not a general result; this is an example in which it holds.